

Increasing Subsequence II

Time limit: 1.00 s

Memory limit: 512 MB

Given an array of n integers, your task is to calculate the number of increasing subsequences it contains. If two subsequences have the same values but in different positions in the array, they are counted separately.

Input

The first input line has an integer n : the size of the array.

The second line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print one integer: the number of increasing subsequences modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$

Example

Input	Output
3 2 1 3	5

Explanation: The increasing subsequences are $[2]$, $[1]$, $[3]$, $[2, 3]$ and $[1, 3]$.